

7 Inch Countertop Touch Screen Kit

Turnkey solution for local touchscreen control of Avi-on lighting control systems

High-Resolution

High-brightness WVGA LCD with integrated multi-touch capacitive touch screen

Adjustable Mounting

Adjustable countertop tilt, or mount inverted under a cabinet

Customizable Interface

Design the screen layout, configure the controller through the RS-232 Interface, and configure to control third-party AV devices

Control Unit & RS-232 Interface

The full range of Avi-on two-way control commands is supported via the RS-232 interface



The Avi-on® 7" Tabletop Touch Screen Kit provides a turnkey solution for touch-screen control of Avi-on lighting control systems. The touch screen is **fully customizable** to support any desired layout. Includes a pre-integrated tabletop mounting base.

Communication to the Avi-on network is achieved through a system controller and the Avi-on RS-232 Serial Interface. **The full range of two-way Avi-on control commands is supported** through the RS-232 interface. Available in Black.

PRODUCT OVERVIEW

Installation

Place the unit on the table and connect a PoE Ethernet cable to the controller. Mount the controller in a suitable location with reliable access to the Avi-on mesh network (near lighting controllers or power packs). Connect an Ethernet cable between the touch screen and the controller using a router and wire the RS-232 Interface to the controller. Apply power to the controller and provide 12 VDC to the RS-232 Interface.

Note: A router is required to assign an IP to the touch screen.

Programming

Design the screen layout using the provided Windows-based software toolkit, and configure the controller interface to communicate through the RS-232 Interface (standard serial configuration).

Program the control commands using the Avi-on Serial Control Protocol documentation. Some advanced control actions may require additional configuration on the Avi-on network. Programming services for lighting control applications are available from Avi-on.

Applications

Ideal for conference rooms, control rooms, stadium command centers, and any location requiring a locally communicating, customizable touch-screen interface.

The touch screen can also be configured (by others) to control a wide range of third-party AV devices.

ORDERING INFORMATION

	Part Number	Description	Application	Color
☐	AVI-RTI-CX7	7" Touch Screen	Under-Cabinet or Counter Control	Black
☐	AVI-KIT-RTI-XP3	Touch Screen Control Unit and RS232 Interface	Touch Screen Controller	-

Project		Location/ Type	
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SPECIFICATIONS

Touch Screen

Power:	(Power supply not included) +12 VDC, 2A, or PoE (802.3af)
Power Management:	Adjustable backlight / proximity sensor
Touchscreen Display:	High-brightness color IPS LCD, FWWGA (854 × 480 pixels) with true-black display filter, Integrated high-resolution capacitive-touch screen
Display Size:	8" (203 mm) diagonal, 16:10 aspect ratio
Display Backlight:	LED backlit, 400 nit
Ethernet Port:	RJ45, 10/100/1000 Base-T
System Memory:	256 MB Flash (non-volatile) memory
Macro Capability:	Total number of macros limited only by available memory
Operating Temperature:	+32°F to +122°F (0°C to +50°C)
Operating Humidity:	0.5% to 95% non-condensing
Mounting:	Tabletop, or screws through mounting bracket (included)
Installation:	U.S. standard single or dual gang
Dimensions:	7.5" × 7.3" × 6.25" (W × H × D) (190 mm × 185 mm × 159 mm)
Weight:	2.0 lbs (907 g)

Control Processor

Power Supply:	+12 VDC, 1A, or PoE (802.3af)
Multi-Purpose Output Port:	(Port 1) One custom 3.5 mm, 4-conductor jack. Designed for RTI modules as well as industry-standard IR emitters.
IR Output Ports:	(Ports 2 & 3) Two 3.5 mm jacks: Compatible with industry-standard IR emitters.
IR Output Drive:	100 mA maximum (per port, adjustable) Infrared
Frequency Transmission Range:	15 kHz – 460 kHz
RS-232 Port:	One, bi-directional
Ethernet Port:	RJ45, One 10/100Base-T with Power-over-Ethernet (Class 3)
Relays:	Two, 5 Amps @ 30 VDC or 12 V / 100 mA
Trigger Sense Inputs:	Two, 3–24 VDC, Off/Low voltage condition must not exceed 1 VDC
USB Ports:	One
Mounting:	Wall-mount or free-standing
Operating Temperature:	+32°F to +122°F (0°C to +50°C)
Operating Humidity:	0.5% to 95% non-condensing
Dimensions (W × H × D):	7.0" × 3.0" × 1.0" (177 mm × 76 mm × 25 mm)
Weight:	6.4 oz (100 g)

RS-232 Serial Interface

Input Voltage:	12–24 VDC
Power Consumption:	17 mA
CC Input:	12–24 VDC non-powered contact closure detection (momentary)
Direct Connect Input:	Compatible with all Avi-on Direct Connect™ Motion/Daylight Sensors
Size:	2.30 in × 1.43 in × 0.75 in (58.2 mm × 36.4 mm × 19 mm)
Mounting:	Removable mounting tabs
Weight:	0.45 oz (16 g)
Terminal Blocks:	22–16 AWG wires
Operating Temperature:	–22°F to +158°F (–30°C to +70°C)
Storage Temperature:	–40°F to +185°F (–40°C to +85°C)
Humidity Rating:	95% non-condensing
Radio Frequency:	2.4 GHz
Wireless Standard:	BLE 5.3 with Mesh
Point-to-Point Range*:	80 ft with obstructions; 350 ft unobstructed
Security:	AES 128-bit encryption for device-to-device communication AES 256-bit encryption for device-to-cloud communication
Warranty:	5 years; 10 years optional
Regulatory:	FCC ID: 2AFZI-AVIBG21 IC ID: 20544-AVIBG21 BQB: DID: D063032 CE: EN IEC 61000-6-1:2019, EN IEC 61000-6-3:2021

*When communicating through the mesh, range is essentially unlimited (5000ft+)

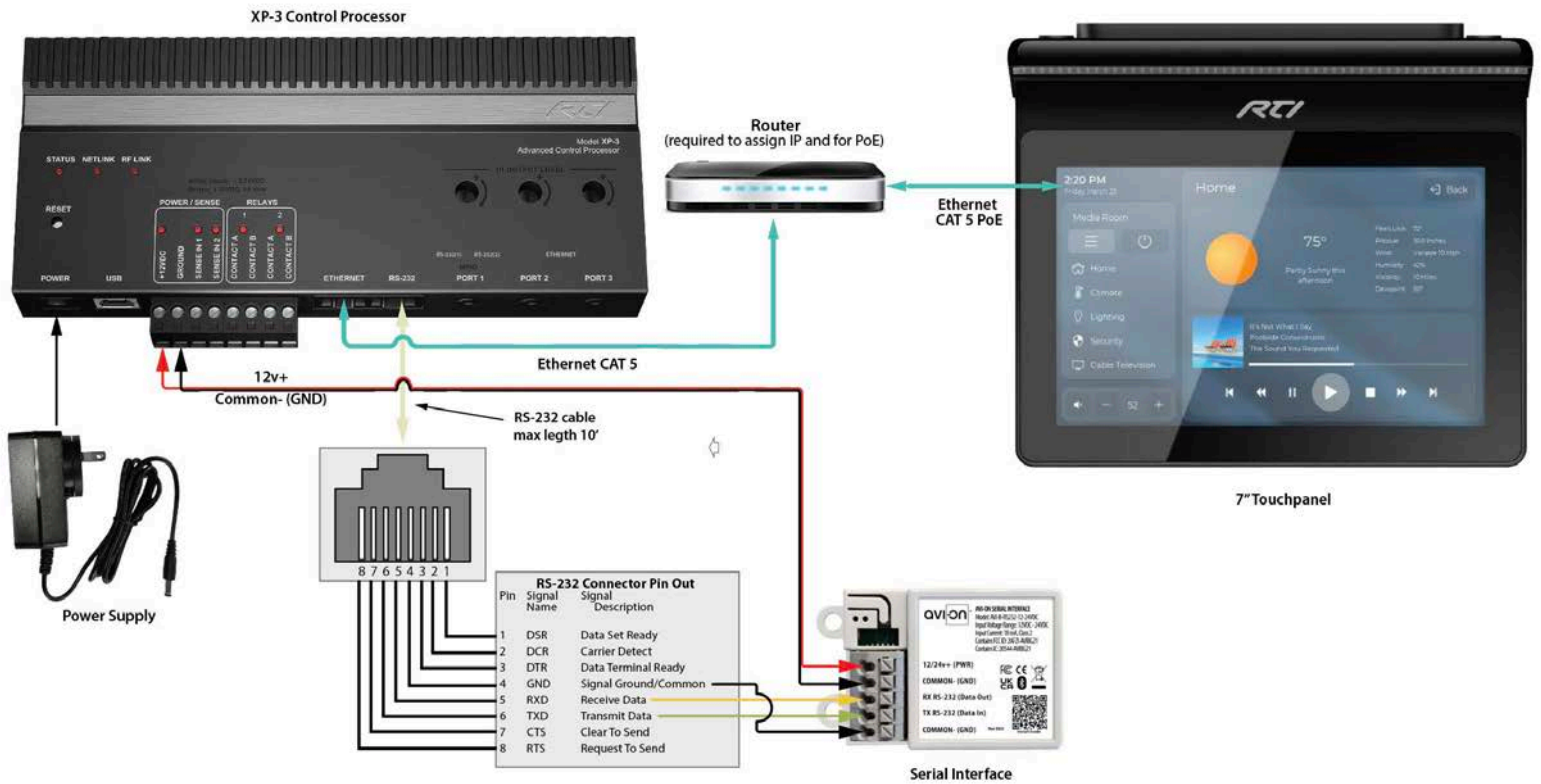
ADDITIONAL IMAGES



Countertop vs Under-Cabinet Mounting
AVI-RTI-CX7

**Integrated Central Processor +
Serial Interface**
AVI-KIT-RTI-XP3

WIRING



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